

Assessing Changes to California Wildfire Causes and Burn Areas Over Time

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2026-04-24

Contents

Rationale and Introductions	2
Research Questions	2
Data Information	2
Exploratory Data Analysis	4
Change Over Time	4
Fire Causes	6
Mapping Burn Areas	9
Analysis	10
Question 1: How has the size of wildfire burn areas of changed over time?	10
Burn Area Yearly Time Series	11
Mann-Kendall	12
Linear Regression	12
Burn Area Monthly Time Series	13
Decomposition	14
Seasonal Mann-Kendall	15
Summary	15
Question 2: How has the number of wildfires caused by lightning strikes changed over time?	16
Lightning-Caused Fires Yearly Time Series	17
Mann-Kendall	17
Lightning-Caused Fires Monthly Time Series	18
Decomposition	19
Seasonal Mann-Kendall	20
Summary	20

Conclusions	21
Limitations	21
References	21

List of Figures

1	Number of Fires Over Time	4
2	Burn Size (Acres) by Year	5
3	Fires Started by Each Cause, 1950 - 2025	6
4	Total Fires Started by Each Cause	7
5	Top Five Fire Causes Over Time	8
6	Map of California Wildfires, 1950 - 2025	9
7	Time Series Decomposition: Monthly Burn Acreage Over Time	14
8	Time Series Plot: Monthly Fires Caused by Lightning Over Time	18
9	Time Series Decomposition: Monthly Count of Fires Caused by Lightning	19

Rationale and Introductions

Wildfires present a pressing environmental challenge in California that has been exacerbated by decades of fire suppression, increasing drought conditions, and a changing climate. This report analyzes wildfire data from the past 75 years, focusing on how climate change has impacted wildfires. Increasingly intense droughts may have a profound affect on Californian fire ecology, contributing to greater fire burn areas, and increased frequency of storms may be a driving higher frequencies of fires caused by lightning striking dry fuels. By providing statistical evidence that wildfire and lightning strike intensity has increased over time, this research can inform policy recommendations, ideally encouraging more proactive fire management strategies to mitigate threats to Californian landscapes.

Research Questions

This project explores the following research questions:

1. How has the size of wildfire burn areas of changed over time?
2. How has the number of wildfires caused by lightning strikes changed over time?

Data Information

This dataset is sourced from the California Natural Resources Agency GIS database and is serviced by the California Department of Forestry and Fire Protection's CAL FIRE eGIS. It includes layers that subset from the larger dataset according to year the fire occurred, and size of the burn area in acres. It contains 17,540 records, spanning from the 1950s to May 9, 2025. It is a publicly available feature layer and was last updated on May 9, 2025.

The following is a summary of the data's column names:

##	[1]	"OBJECTID"	"Year"
##	[3]	"State"	"Agency"
##	[5]	"Unit.ID"	"Fire.Name"
##	[7]	"Local.Incident.Number"	"Alarm.Date"
##	[9]	"Containment.Date"	"Cause"
##	[11]	"Collection.Method"	"Management.Objective"
##	[13]	"GIS.Calculated.Acres"	"Comments"
##	[15]	"Complex.Name"	"IRWIN.ID"
##	[17]	"Fire.Number..historical.use."	"Complex.ID"
##	[19]	"DECADES"	"Shape__Area"
##	[21]	"Shape__Length"	

Exploratory Data Analysis

Exploratory analyses were conducted in order to highlight possible trends or patterns in the data. The following plots reveal how California wildfire sizes and frequency have shifted over the past 75 years, along with leading fire causes.

Change Over Time

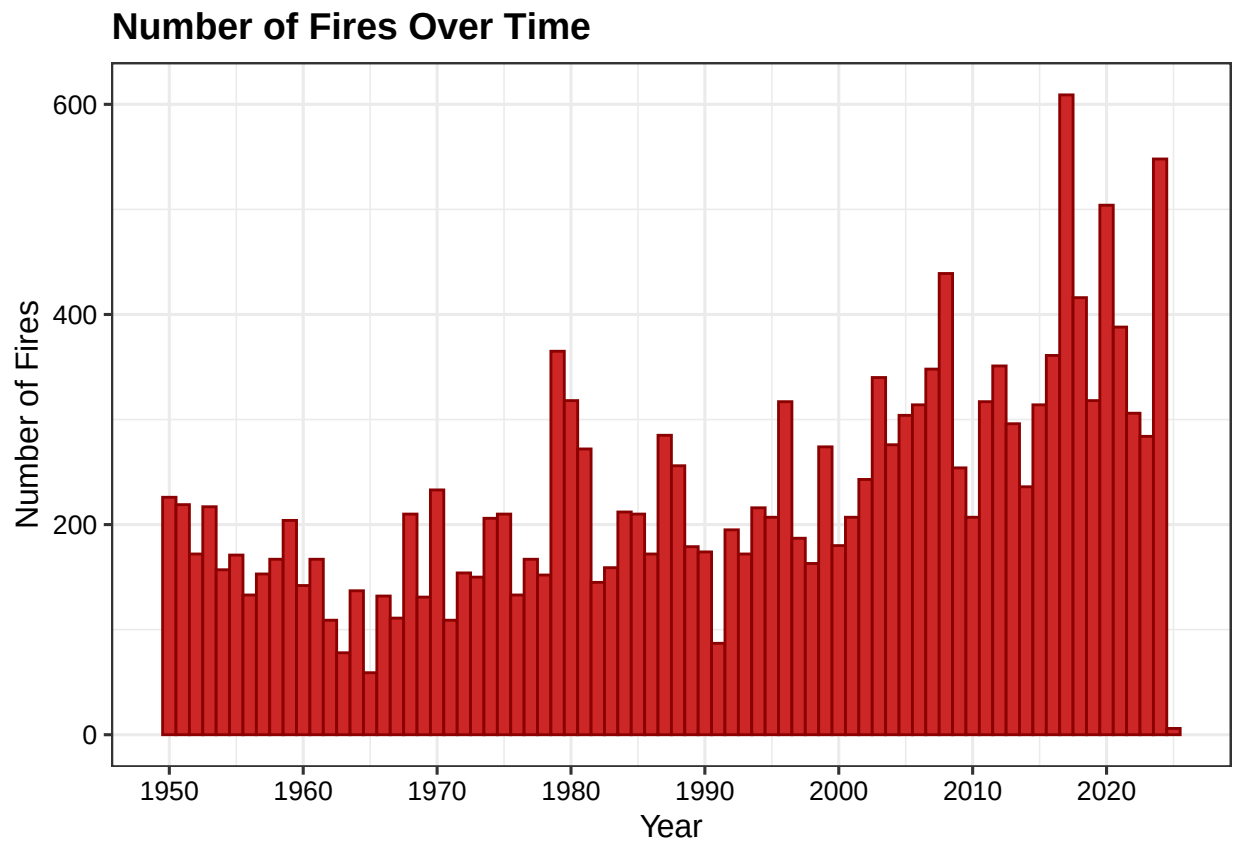


Figure 1: Number of Fires Over Time

This chart plots the number of fires per year for the past 75 years, indicating a possible upward trend in the number of fires over time, with some notable spikes in years when the state exhibited unusually low or high fire activity.

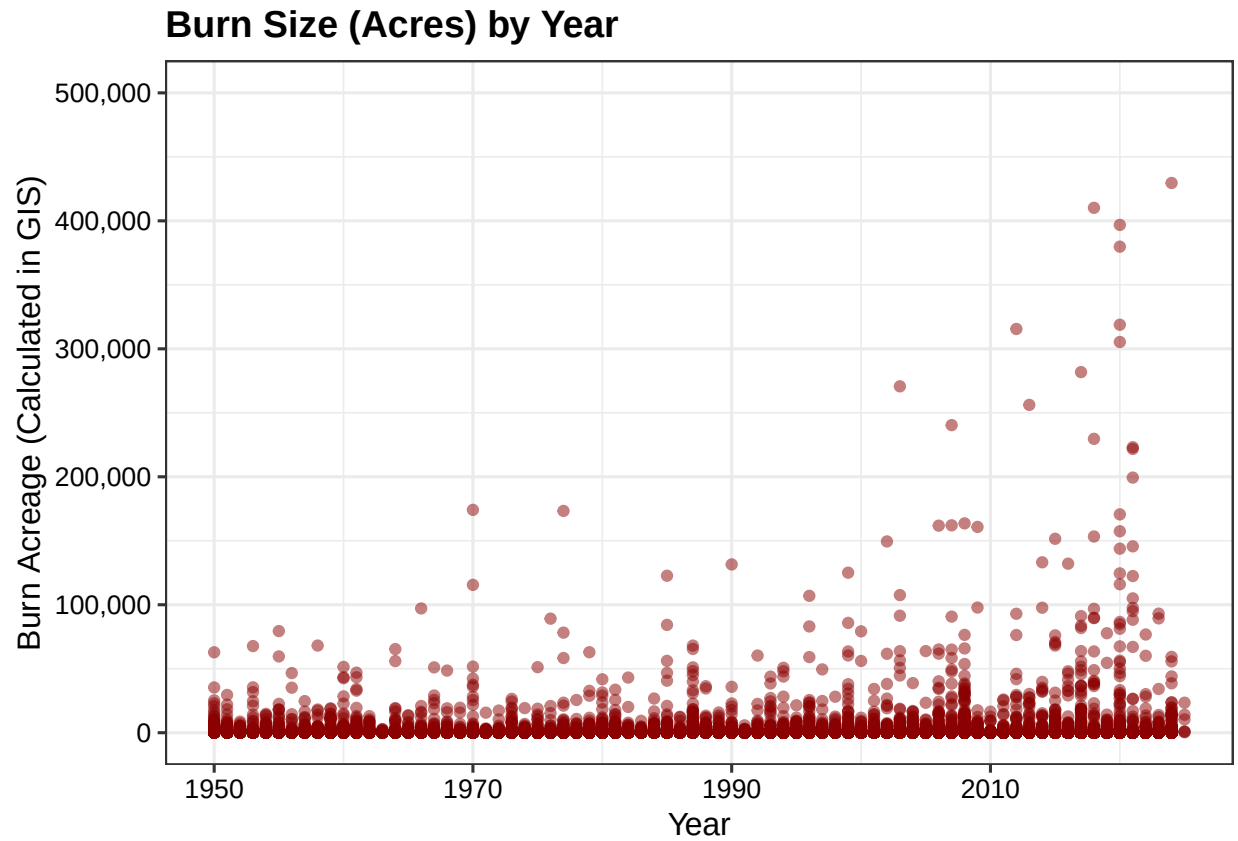


Figure 2: Burn Size (Acres) by Year

This chart plots burn acreage over time, with the removal of a few outlier points. Though the majority of fires covered less than 50,000 acres, there appears to be an increase in points within the 300,000 to 500,000 burn acreage range since 1990. This is an indicator that California's wildfire burn acreage may be increasing over time.

Fire Causes

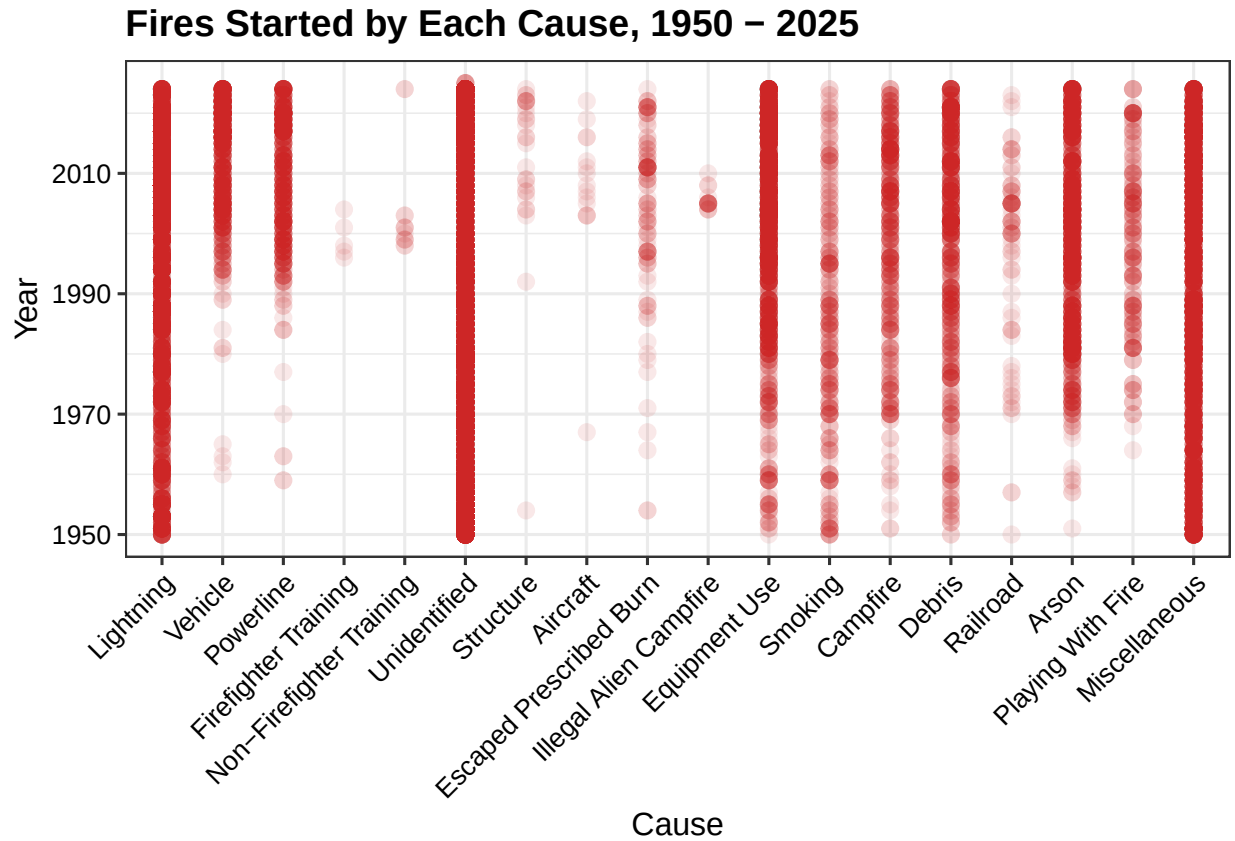


Figure 3: Fires Started by Each Cause, 1950 - 2025

This chart plots wildfire events by ignition cause over time, with each point representing a single fire from 1950–2025. Dense vertical groupings highlight causes that occur more frequently, while sparser columns represent less common ignition sources. Causes that span the full time range appear consistent across decades (Debris and Equipment Use), whereas those concentrated in certain periods (Vehicles and Powerlines) suggest more episodic occurrence or changing patterns over time.

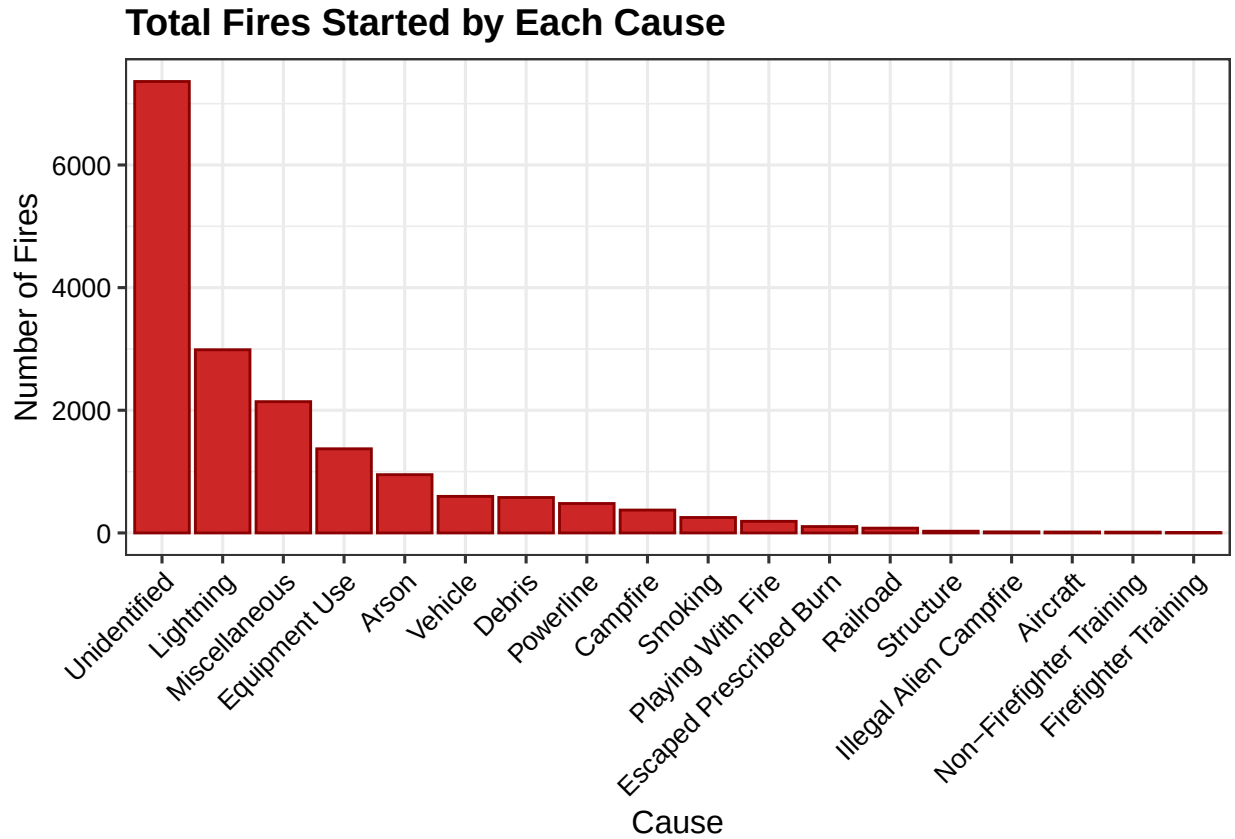


Figure 4: Total Fires Started by Each Cause

This plot reveals that the top five most commonly observed fire causes are:

1. Unidentified
2. Lightning
3. Miscellaneous
4. Equipment Use
5. Arson

Although the metadata does not describe what is meant by Unidentified or Miscellaneous, these categories are too large to ignore. These categories have likely persisted as a result of the vast variety of circumstances that may cause a fire to ignite, along with insufficient resources to conduct a robust investigation surrounding the cause of every small fire.

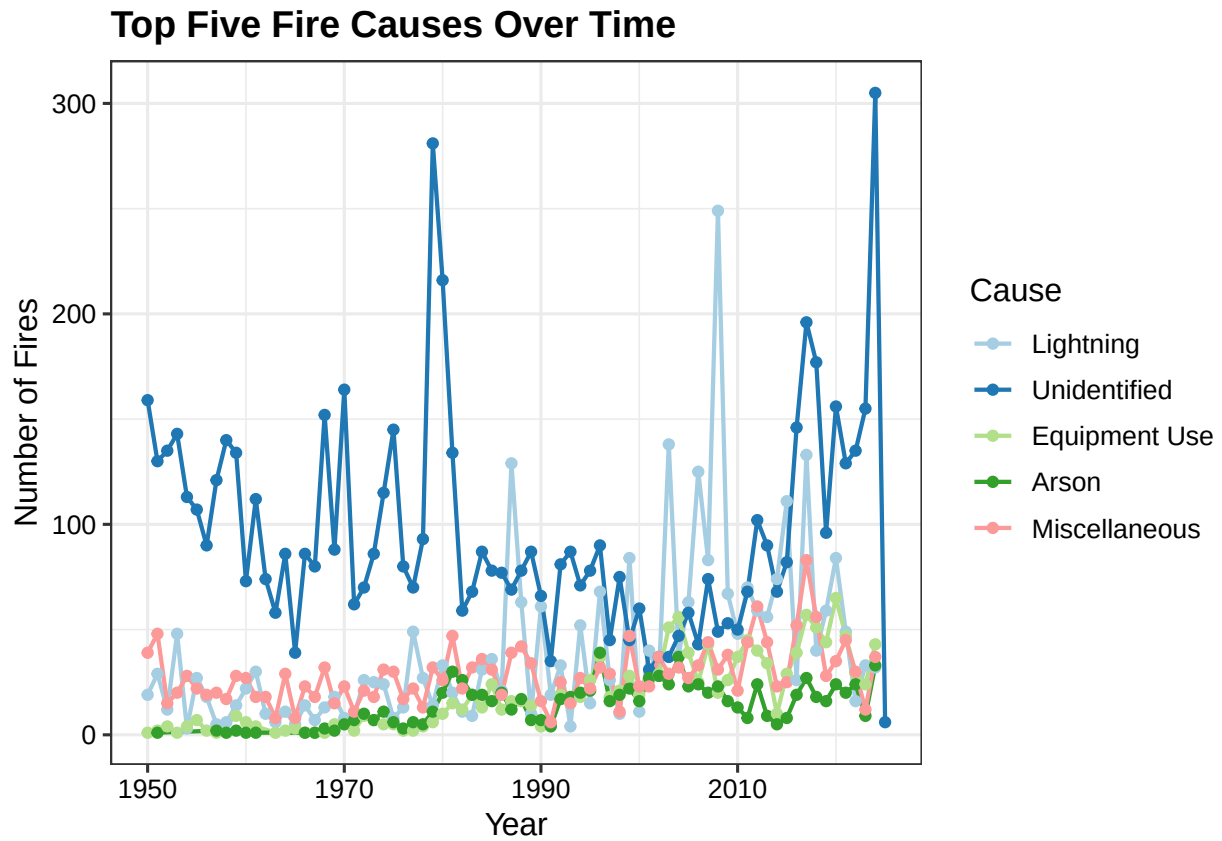


Figure 5: Top Five Fire Causes Over Time

This plot notably reveals that there appears to be an overall increasing trend in the fires caused by lightning, which is the second leading cause of fires in California. Trend analysis for lightning-caused fires is conducted later in this report.

Mapping Burn Areas

Map of California Wildfires, 1950 – 2025

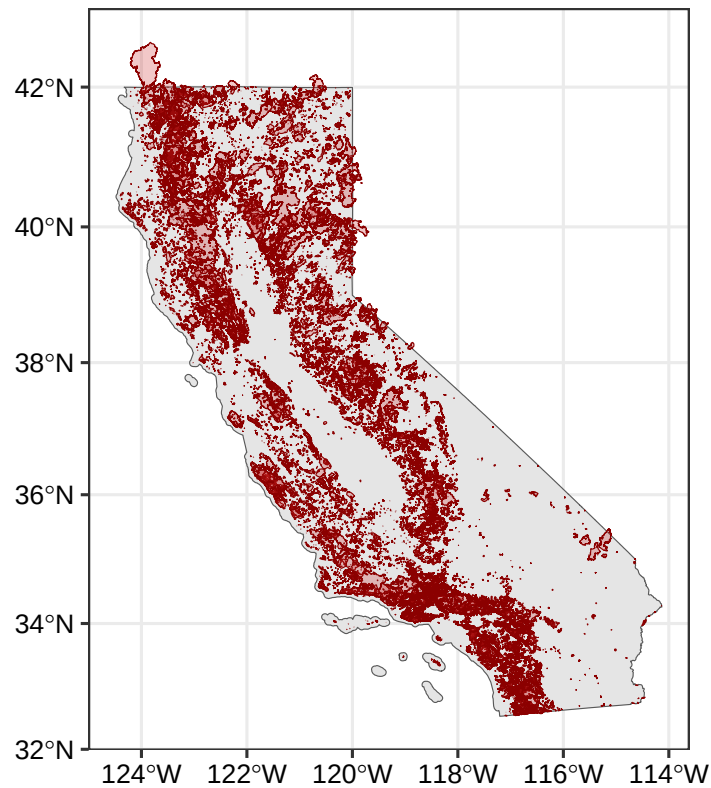


Figure 6: Map of California Wildfires, 1950 - 2025

This map provides an overview of all fire burn areas for the past 75 years. Areas in darker red indicate fire burn area overlap, thus highlighting regions that have been burned by multiple fires throughout the years. Burn areas are notably concentrated around mountain ranges, for the most part excluding California's central valley. This map is consistent with typical fire behavior, providing a visual depiction of how fire always moves uphill.

Analysis

These analyses evaluate how wildfire activity has changed over time, focusing on total burned area and fires caused by lightning. For the burn acreage analysis, fire burn acreage was summed in order to provide monthly and yearly totals. The data was then analyzed at both scales to see if any change over time could be attributed to seasonal patterns.

For the yearly data, a Mann-Kendall test was conducted in order to detect the presence of monotonic trends. For the monthly data, a time series decomposition was used to analyze the data by trend, seasonal patterns, and random variation. A Seasonal Mann-Kendall test was run in order to identify seasonal trends in the monthly data.

The same approach was used for lightning-caused fires, for which monthly and yearly fire counts were summed and analyzed over time. The yearly data was evaluated using a Mann-Kendall test, since there is no seasonal pattern at that scale. Monthly data was again broken down using decomposition, and a Seasonal Mann-Kendall test was used to detect trends while accounting for seasonal changes.

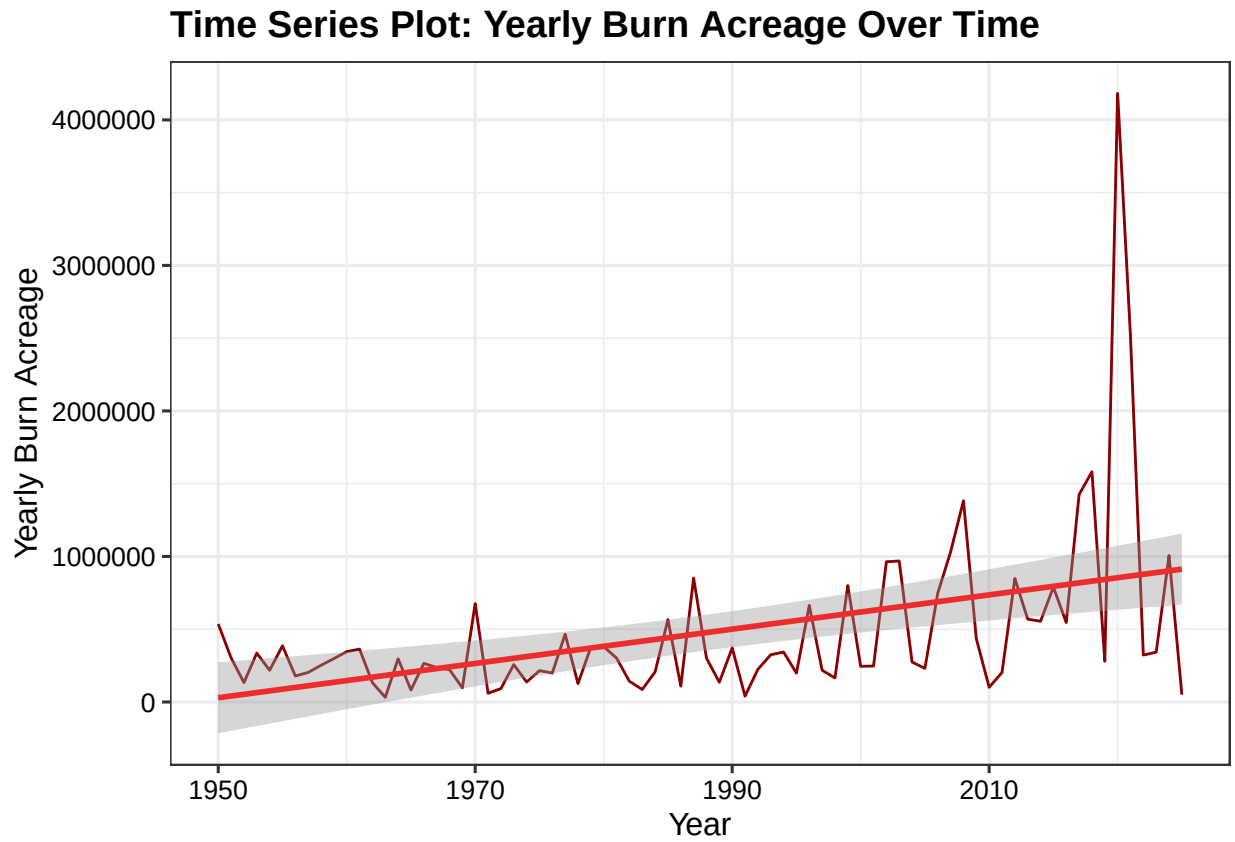
Question 1: How has the size of wildfire burn areas of changed over time?

Null hypothesis: There has been no significant change in burn acreage of California wildfires over time.

Alternative Hypothesis: The burn acreage of California wildfires has significantly increased over time.

The first analysis focuses on whether fire acreage in California has increased, either as a result of an increase in overall fires or an increase in geographically larger fires. A time series was created for the yearly burn acreage over time and then plotted. Then, a Mann Kendall test and linear regression were run in order to identify any trend in the yearly data. Finally, fire acreage was calculated per month to create a time series analysis focused on seasonal trends.

Burn Area Yearly Time Series



This chart plots yearly fire burn acreage over time. The linear trend line notably reveals an upward slope, indicating a strong likelihood of a statistically significant upward trend in burn acreage over time.

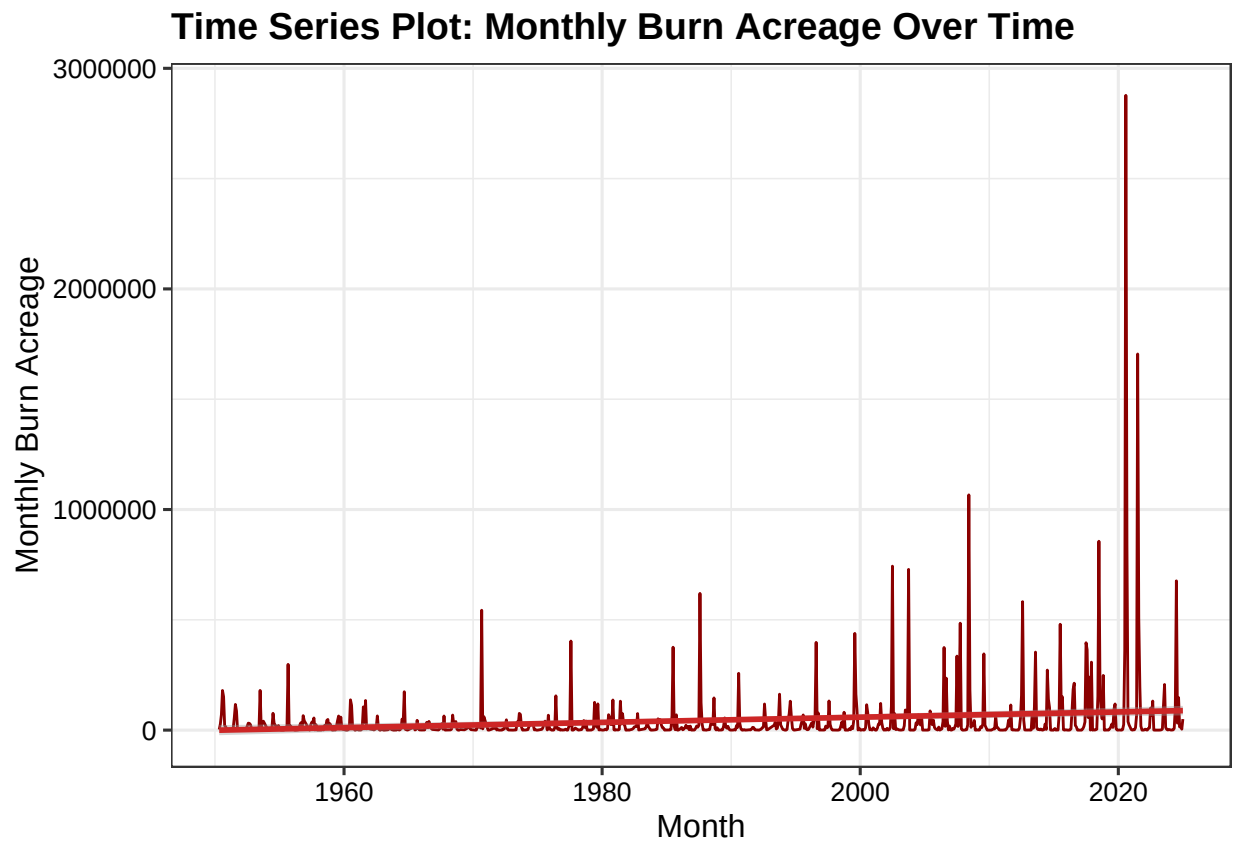
Mann-Kendall

```
## Score = 792 , Var(Score) = 49716.67
## denominator = 2850
## tau = 0.278, 2-sided pvalue =0.00038886
```

Linear Regression

```
##
## Call:
## tslm(formula = fire_acres_ts ~ trend)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -862725 -269881  -67814   132051  3328372
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    17505     124780   0.140    0.889
## trend           11780       2816   4.183 0.0000781 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 538500 on 74 degrees of freedom
## Multiple R-squared:  0.1913, Adjusted R-squared:  0.1803
## F-statistic: 17.5 on 1 and 74 DF, p-value: 0.00007806
```

Burn Area Monthly Time Series



This chart plots monthly fire burn acreage over time. The linear trend line shows a slight upward slope, indicating a possibility of a statistically significant upward trend in monthly burn acreage over time.

Decomposition

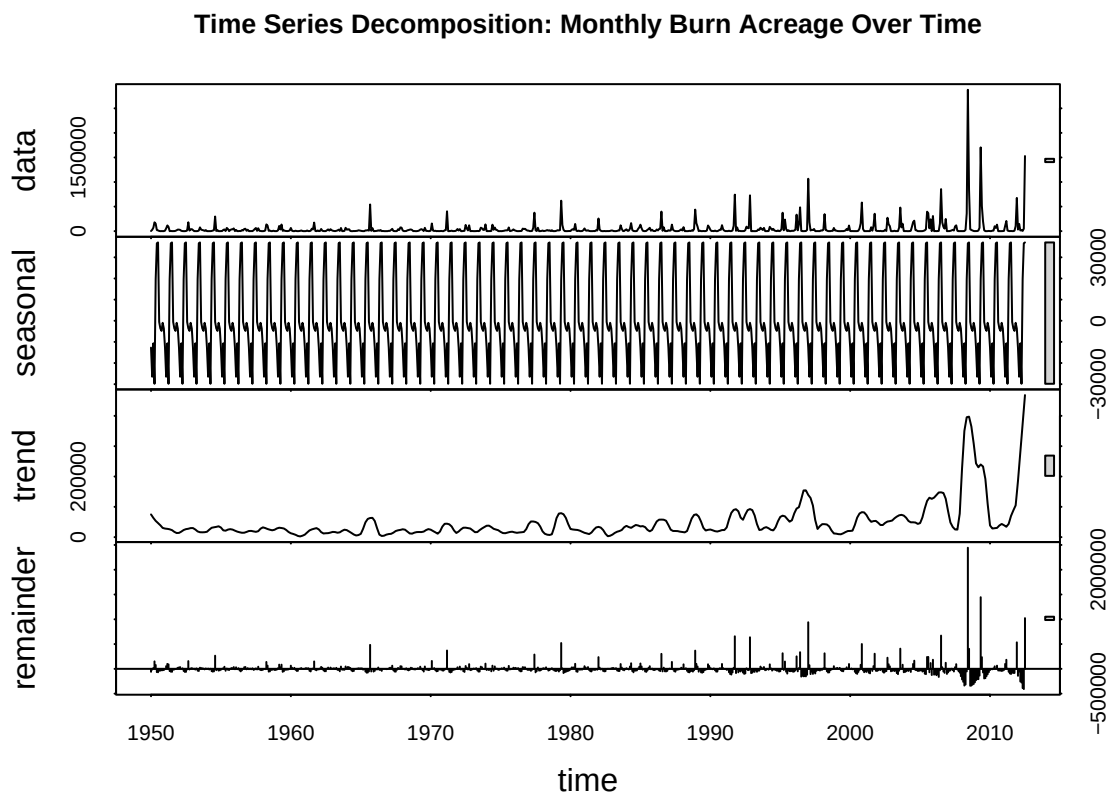


Figure 7: Time Series Decomposition: Monthly Burn Acreage Over Time

Seasonal Mann-Kendall

```
##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: monthly_acres_ts
## z = 1.6477, p-value = 0.09941
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##      S      varS
## 954.0 334510.7

##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: monthly_acres_ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##
##      S      varS      tau      z Pr(>|z|)
## Season 1: S = 0 -55 28427.0 -0.028 -0.320 0.748757
## Season 2: S = 0 -119 28427.0 -0.061 -0.700 0.484009
## Season 3: S = 0 -49 28427.0 -0.025 -0.285 0.775880
## Season 4: S = 0 -21 28427.0 -0.011 -0.119 0.905575
## Season 5: S = 0 189 28427.0 0.097 1.115 0.264831
## Season 6: S = 0 221 28427.0 0.113 1.305 0.191947
## Season 7: S = 0 289 28427.0 0.148 1.708 0.087608 .
## Season 8: S = 0 139 27104.3 0.074 0.838 0.401905
## Season 9: S = 0 83 27104.3 0.044 0.498 0.618431
## Season 10: S = 0 65 27104.3 0.034 0.389 0.697468
## Season 11: S = 0 187 27104.3 0.099 1.130 0.258569
## Season 12: S = 0 25 27104.3 0.013 0.146 0.884097
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Summary

The Seasonal Mann-Kendall test of monthly fire burn acreage demonstrates variability and occasional increases in wildfire activity, but no significant trend in fire burn area over time at the monthly scale once seasonality is accounted for. The monthly time series decomposition supports this: the trend component shows a gradual upward slope with occasional sharp jumps that reflect extreme fire years and major shifts in conditions, but the seasonal component shows a stable, repeating annual fire cycle, and the remainder plot highlights large spikes in activity that represent unusually severe fire events. This contrasts with the yearly fire acreage trend discussed below.

The yearly fire burn area Mann-Kendall test and linear regression show a clear, statistically significant increase in burn acreage over time, with a positive coefficient denoting the increasing trend. The R^2 value of 0.1803 means that about 18% of the change in burn area sizes over time can be explained by time alone ($p < 0.05$). These results confirm the alternative hypothesis that the burn acreage of California wildfires has significantly increased over time. Overall, the combined monthly and yearly results suggest that wildfire sizes are increasing over time, but much of the change is driven by long-term shifts and extreme fire events rather than a steady month-to-month increase.

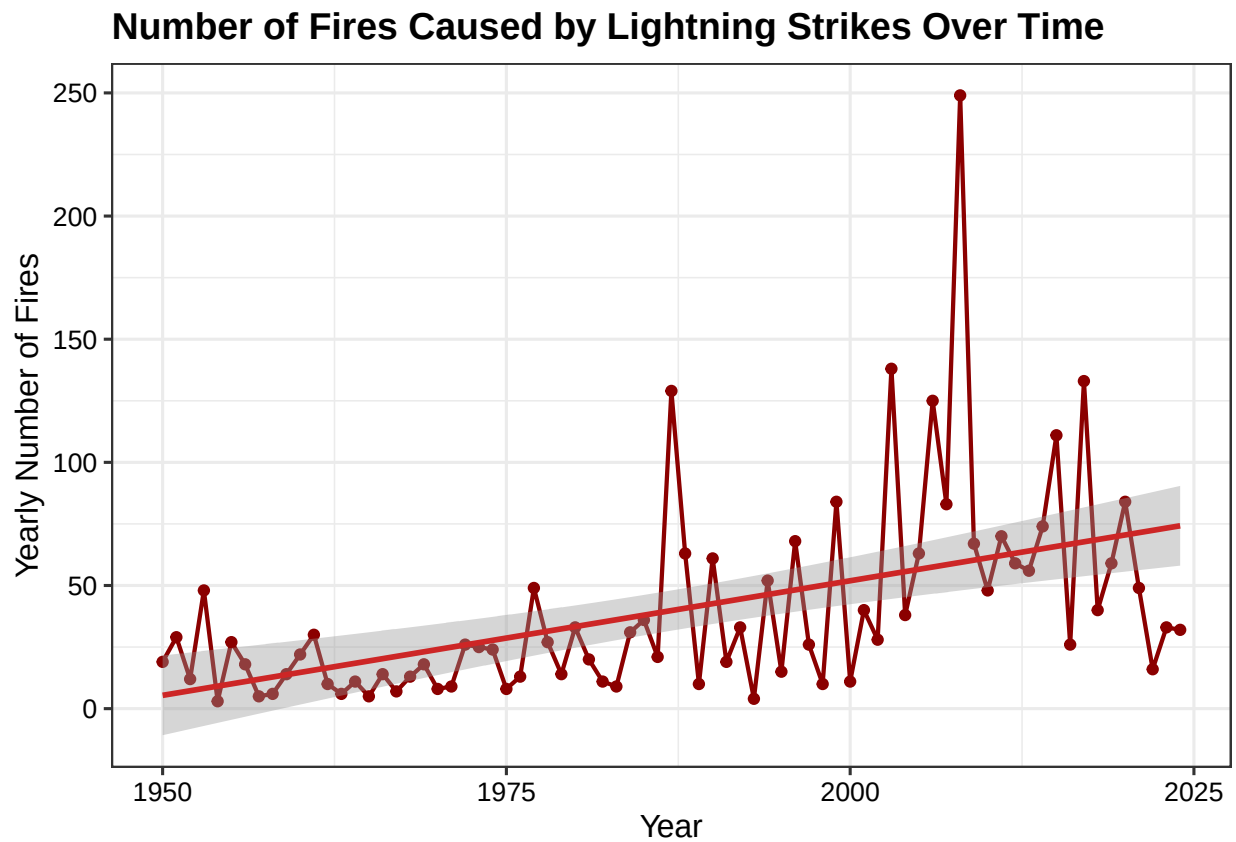
Question 2: How has the number of wildfires caused by lightning strikes changed over time?

Null hypothesis: There has been no significant change in the number of California wildfires caused by lightning over time.

Alternative Hypothesis: The number of California wildfires caused by lightning strikes has significantly increased over time

Informed by the exploratory analysis, these time series investigate whether there is sufficient statistical evidence to determine if the number of fires caused by lightning strikes has increased over time. A time series was created for the number of lightning-caused fires each year and then plotted. Then, a Mann Kendall test was run in order to identify any trend in the yearly data. Finally, the number of lightning-caused fires per month was calculated in order to create a time series analysis that breaks down seasonal trends.

Lightning-Caused Fires Yearly Time Series



This chart plots the yearly number of fires caused by lightning over time. The linear trend line has a notable upward slope, indicating a possible significant upward trend in lightning-caused fires over time.

Mann-Kendall

```
## Score = 1272 , Var(Score) = 393973.3
## denominator = 11459.99
## tau = 0.111, 2-sided pvalue =0.042874
```

Lightning-Caused Fires Monthly Time Series

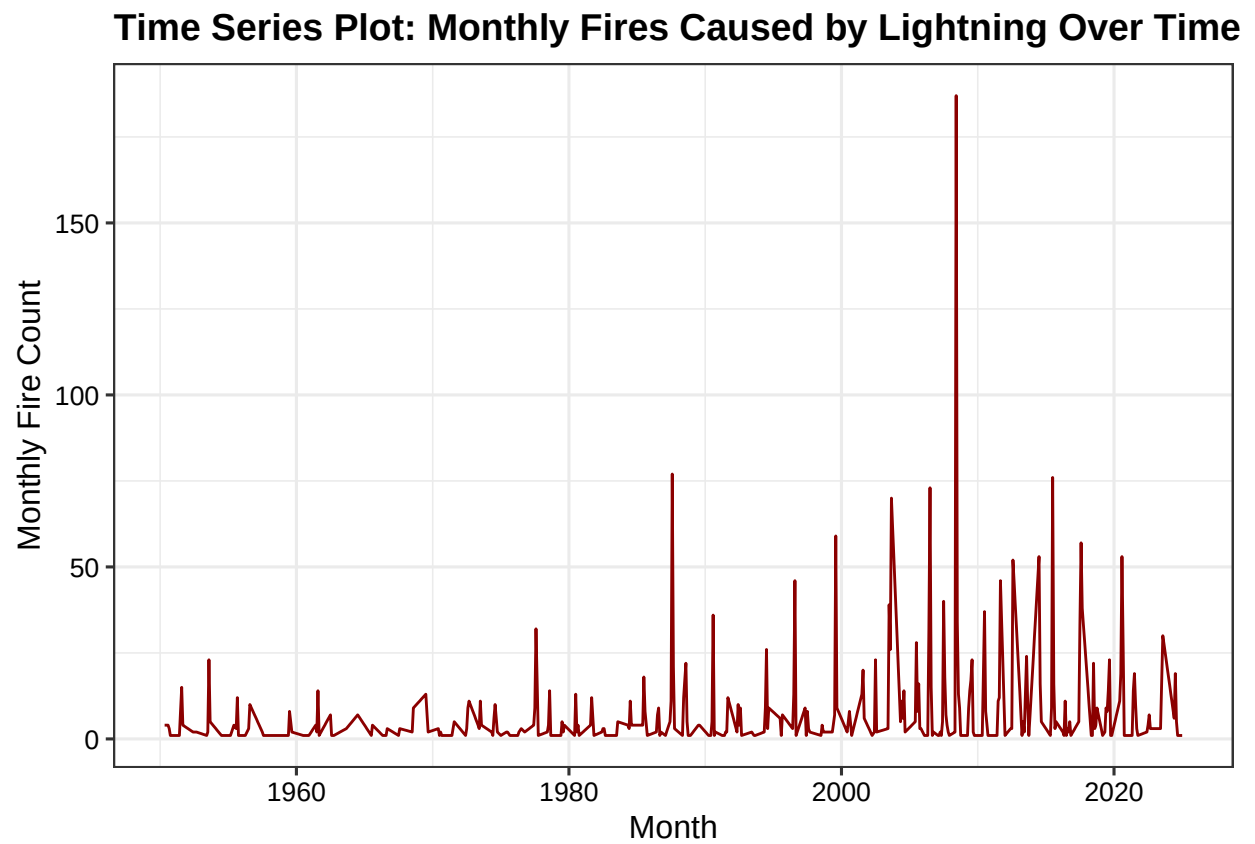


Figure 8: Time Series Plot: Monthly Fires Caused by Lightning Over Time

Decomposition

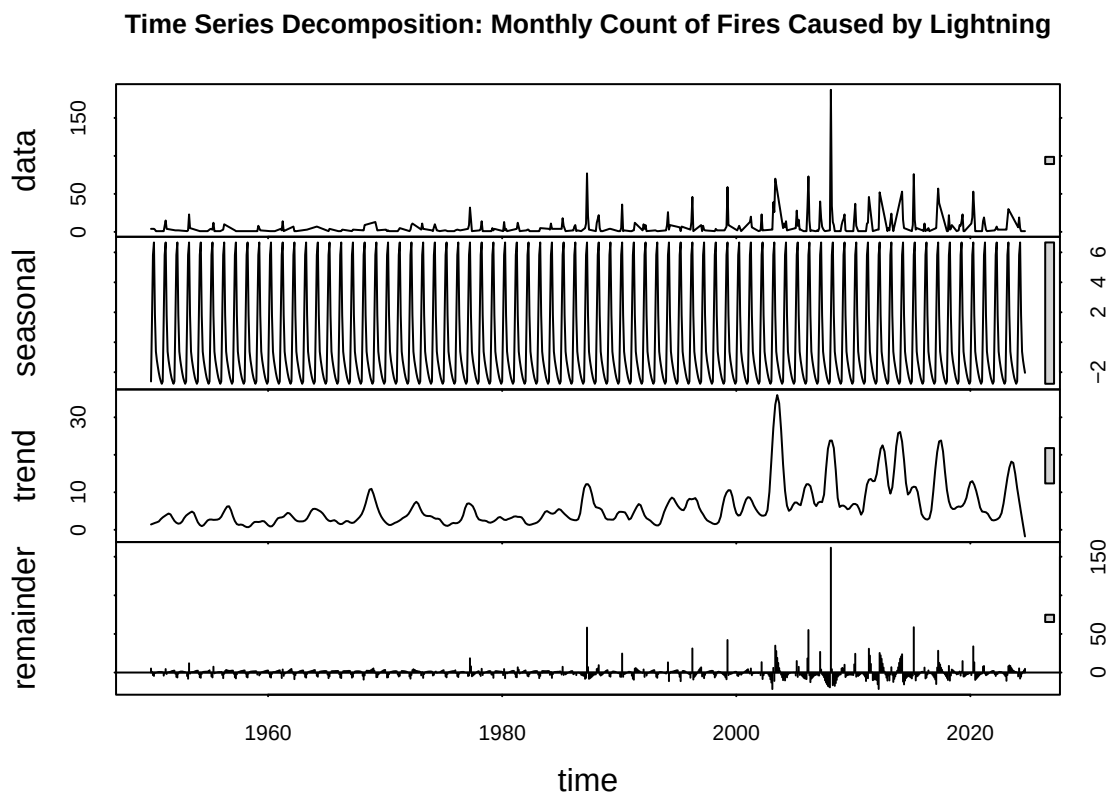


Figure 9: Time Series Decomposition: Monthly Count of Fires Caused by Lightning

Seasonal Mann-Kendall

```
##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: lightning_monthly_ts
## z = 7.4159, p-value = 0.0000000000001208
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##      S      varS
## 5550.0 559882.7

##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: lightning_monthly_ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##
##      S      varS      tau      z      Pr(>|z|)
## Season 1: S = 0 182 47003.0 0.068 0.835 0.40379423
## Season 2: S = 0 793 47094.3 0.297 3.650 0.00026269 ***
## Season 3: S = 0 1170 47522.7 0.432 5.362 0.000000082096 ***
## Season 4: S = 0 822 47568.0 0.303 3.764 0.00016701 ***
## Season 5: S = 0 536 47072.7 0.201 2.466 0.01366827 *
## Season 6: S = 0 302 46115.3 0.115 1.402 0.16101570
## Season 7: S = 0 291 46956.3 0.109 1.338 0.18080124
## Season 8: S = 0 328 47183.3 0.122 1.505 0.13221991
## Season 9: S = 0 263 47281.3 0.097 1.205 0.22823609
## Season 10: S = 0 341 45383.0 0.130 1.596 0.11048926
## Season 11: S = 0 290 45397.0 0.110 1.356 0.17497542
## Season 12: S = 0 232 45305.7 0.088 1.085 0.27780439
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Summary

The Mann-Kendall trend test found that there has been a significant increase in the annual number of lightning-caused fires over time ($\tau = 0.111$, $p = 0.043$). Moreover, the Seasonal Mann-Kendall test reveals that the monthly number of lightning-caused fires has also increased over time, even when normal seasonal patterns are taken into account ($z = 7.42$, $p < 0.001$). The increase was strongest during late winter and spring, especially in Seasons 2 through 5 (February - May), with Season 3 (March) showing the largest rise over time.

The decomposition of the monthly time series shows that lightning-caused fires follow a regular yearly pattern, with more fires in some months and fewer in others. This means the timing of these fires is fairly consistent from year to year, following a seasonal pattern. It also shows that the overall number of lightning-caused fires has gradually increased over time, with more recent years generally having higher counts. The ups and downs of the remainder plot represent unusual months with especially high or low fire activity that are not consistent with the typical seasonal pattern.

Overall, these results demonstrate that the number of California wildfires caused by lightning strikes has significantly increased over time at both a monthly and yearly scale, thus confirming the alternative hypothesis.

Conclusions

This report found that the size of California wildfires has significantly increased over time, and that the number of wildfires caused by lightning strikes has also significantly increased over time, at both a monthly and yearly scale. Results of the time series decomposition point to increasingly extreme fire events as a possible driver of the increase in burn acreage. Further analysis investigating the causes of these larger, extreme fire events may be useful in developing a more in depth understanding of the trend. Additional analyses investigating whether the increase in lightning-caused fires can be attributed to a greater frequency of electrical storms or higher quantities of dry fire fuels may prove extremely useful for fire management strategy, and can help empirically link these observed trends to global climate change. Overall, the trends identified in this report paint a picture of the rapidly shifting Californian fire ecology, highlighting the need for proactive, scientifically-backed management strategies to mitigate the effects of a changing climate and protect diverse taxa, ecosystems, and ways of life.

Limitations

It is noted in the metadata for this dataset that there are a number of fires missing from these records, mostly during the first few decades of the data. As a result, certain trend analyses may present slightly differently than they would if every fire from the last 75 years was able to be included. This is important to note when drawing conclusions from the data, particularly when focusing the on yearly and monthly fire acreage.

The two categories of “Unidentified” and “Miscellaneous” also present a degree of limitation due to a lack of specificity. It would be interesting to run the exploratory analyses again if the fires in these categories were able to be grouped correctly by their specific causes.

References

Github Repository: https://github.com/skyvolz/SkyVolz_BriannaKarson

Fire Data:

URL: <https://gis.data.cnra.ca.gov/datasets/CALFIRE-Forestry::california-fire-perimeters-1950/about>

Citation Title: Fire Perimeters (firep24_1)

Revision date: 2025-04-23

Edition: 24_1

Processing environment Version 6.2 (Build 9200); Esri ArcGIS 10.8.0.12790

CA Boundary Shapefile:

URL: https://data.ca.gov/dataset/e212e397-1277-4df3-8c22-40721b095f33/resource/3db1e426-fb51-44f5-82d5-a54d7c6e188b/download/ca_state.zip

California State boundary in shapefile format from the US Census Bureau’s 2023 MAF/TIGER database.